

**Tugas Week 4**  
**Jaringan Komputer**



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## 4.2 Nslookup

### Bagian A – Soal 1–3

#### Soal

1. Jalankan nslookup untuk mendapatkan alamat IP dari server web di Asia. Berapa alamat IP server tersebut?
  2. Jalankan nslookup agar dapat mengetahui server DNS otoritatif untuk universitas di Eropa.
  3. Jalankan nslookup untuk mencari tahu informasi mengenai server email dari Yahoo! Mail melalui salah satu server yang didapatkan di pertanyaan nomor 2. Apa alamat IP-nya?
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#### Jawaban

1. **Alamat IP server web di Asia:**  
45.60.35.225
  2. **Server DNS otoritatif Universitas Oxford (Eropa):**
    1. auth6.dns.ox.ac.uk
    2. dns2.ox.ac.uk
    3. dns0.ox.ac.uk
    4. dns1.ox.ac.uk
    5. auth4.dns.ox.ac.uk
    6. auth5.dns.ox.ac.uk
  3. **Alamat IP server email Yahoo! Mail (melalui salah satu server DNS di atas):**  
129.67.1.190
- 

#### Bukti

- **No 1:**

```
Server: gpon.net
Address: fe80::1

Non-authoritative answer:
Name: mgnzsqc.x.incapdns.net
Address: 45.60.35.225
Aliases: www.nus.edu.sg
```

- **No 2:**

```
Server: gpon.net
Address: fe80::1

Non-authoritative answer:
ox.ac.uk      nameserver = auth6.dns.ox.ac.uk
ox.ac.uk      nameserver = dns2.ox.ac.uk
ox.ac.uk      nameserver = dns0.ox.ac.uk
ox.ac.uk      nameserver = dns1.ox.ac.uk
ox.ac.uk      nameserver = auth4.dns.ox.ac.uk
ox.ac.uk      nameserver = auth5.dns.ox.ac.uk

dns0.ox.ac.uk internet address = 129.67.1.190
dns1.ox.ac.uk internet address = 129.67.1.191
dns2.ox.ac.uk internet address = 163.1.2.190
auth4.dns.ox.ac.uk internet address = 45.33.127.156
auth5.dns.ox.ac.uk internet address = 93.93.128.67
auth6.dns.ox.ac.uk internet address = 185.24.221.32
auth4.dns.ox.ac.uk AAAA IPv6 address = 2600:3c00:e000:19::1
auth5.dns.ox.ac.uk AAAA IPv6 address = 2a00:1098:0:80:1000::10
auth6.dns.ox.ac.uk AAAA IPv6 address = 2a02:2770:11:0:21a:4aff:febe:759b
```

- **No 3:**

```
Server: auth0.dns.ox.ac.uk
Address: 129.67.1.190
```

## Bagian B – Soal 1–7

### Soal

1. Cari pesan permintaan DNS dan balasannya. Apakah pesan tersebut dikirimkan melalui UDP atau TCP?
2. Apa port tujuan pada pesan permintaan DNS? Apa port sumber pada pesan balasannya?
3. Pada pesan permintaan DNS, apa alamat IP tujuannya? Apa alamat IP server DNS lokal anda (gunakan ipconfig untuk mencari tahu)? Apakah kedua alamat IP tersebut sama?
4. Periksa pesan permintaan DNS. Apa “jenis” atau “type” dari pesan tersebut? Apakah pesan permintaan tersebut mengandung “jawaban” atau “answers”?
5. Periksa pesan balasan DNS. Berapa banyak “jawaban” atau “answers” yang terdapat di dalamnya? Apa saja isi yang terkandung dalam setiap jawaban tersebut?
6. Perhatikan paket TCP SYN yang selanjutnya dikirimkan oleh host Anda. Apakah alamat IP pada paket tersebut sesuai dengan alamat IP yang tertera pada pesan balasan DNS?
7. Halaman web yang sebelumnya anda akses (<http://www.ietf.org>) memuat beberapa gambar. Apakah host Anda perlu mengirimkan pesan permintaan DNS baru setiap kali ingin mengakses suatu gambar?

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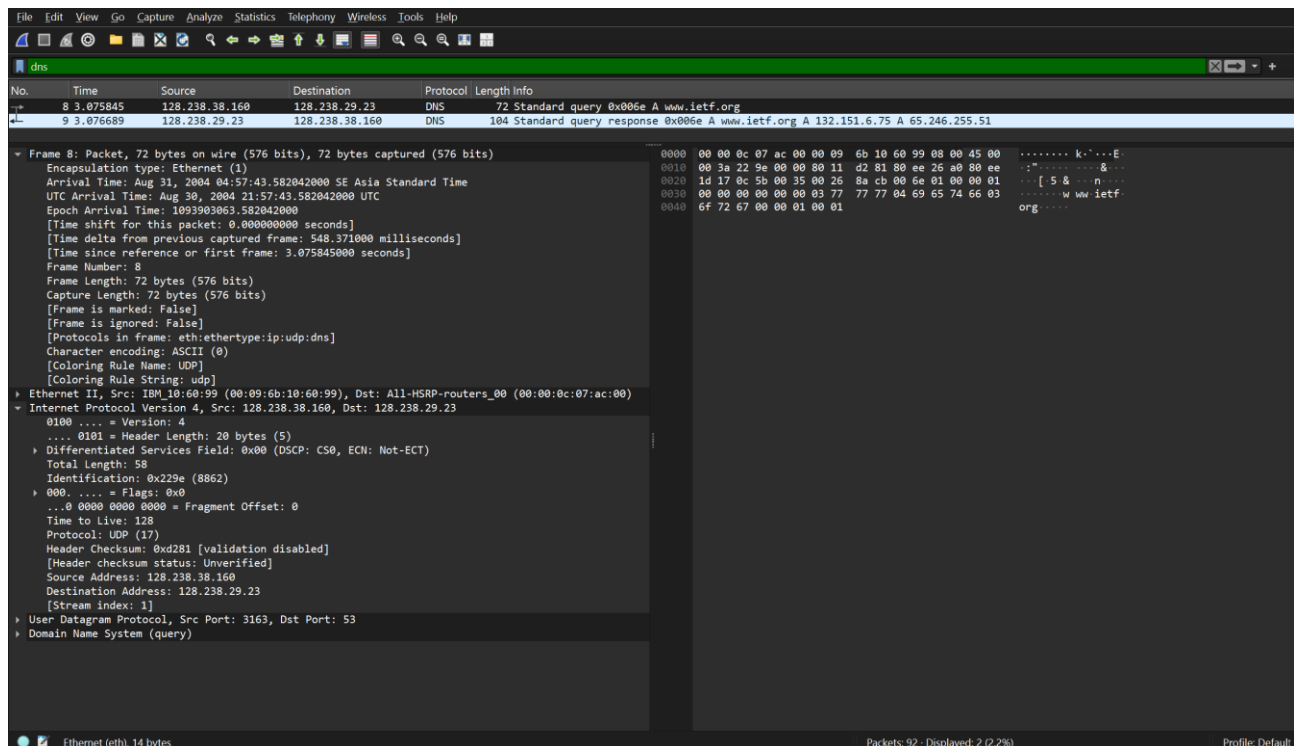
### Jawaban

1. **Protokol:** UDP
2. **Port:**

- Tujuan request: 53
  - Sumber request: 3163
  - Sumber response: 53
3. **Alamat IP:** Sama (IP tujuan = IP server DNS lokal)
4. **Type:** A
- Tidak ada *answers* pada request
5. **Balasan:** 2 *answers*
- www.ietf.org → 132.151.6.75
  - www.ietf.org → 65.246.255.51
6. **Kesesuaian IP:** Ya, sesuai
7. **Permintaan DNS baru:** Tidak

## Bukti

- **Request:**



- **Response:**

| No. | Time     | Source         | Destination    | Protocol | Length | Info                                                                         |
|-----|----------|----------------|----------------|----------|--------|------------------------------------------------------------------------------|
| 8   | 3.075845 | 128.238.38.160 | 128.238.29.23  | DNS      | 72     | Standard query 0x006e A www.ietf.org                                         |
| 9   | 3.076689 | 128.238.29.23  | 128.238.38.160 | DNS      | 104    | Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.255.51 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Frame 9: Packet, 104 bytes on wire (832 bits), 104 bytes captured (832 bits)</p> <p>Encapsulation type: Ethernet (1)</p> <p>Arrival Time: Aug 31, 2004 04:57:43.582886000 SE Asia Standard Time</p> <p>UTC Arrival Time: Aug 30, 2004 21:57:43.582886000 UTC</p> <p>Epoch Arrival Time: 1093903063.582886000</p> <p>[Time shift for this packet: 0.000000000 seconds]</p> <p>[Time delta from previous captured frame: 844.000 microseconds]</p> <p>[Time delta from previous displayed frame: 844.000 microseconds]</p> <p>[Time since reference or first frame: 3.076689000 seconds]</p> <p>Frame Number: 9</p> <p>Frame Length: 104 bytes (832 bits)</p> <p>Capture Length: 104 bytes (832 bits)</p> <p>[Frame is marked: False]</p> <p>[Frame is ignored: False]</p> <p>[Protocols in frame: eth:ethertype:ip:udp:dns]</p> <p>Character encoding: ASCII (0)</p> <p>[Coloring Rule Name: UDP]</p> <p>[Coloring Rule String: udp]</p> <p>Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)</p> <p>Destination: IBM_10:60:99 (00:09:6b:10:60:99)</p> <p>Source: Cisco_83:e4:54 (00:b0:8e:83:e4:54)</p> <p>Type: IPv4 (0x0800)</p> <p>[Stream index: 6]</p> <p>Internet Protocol Version 4, Src: 128.238.29.23, Dst: 128.238.38.160</p> <p>0100 .... = Version: 4</p> <p>.... 0101 = Header Length: 20 bytes (5)</p> <p>Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)</p> <p>Total Length: 90</p> <p>Identification: 0xd595 (54677)</p> <p>0000 .... = Flags: 0x0</p> <p>...0 0000 0000 0000 = Fragment Offset: 0</p> <p>Time to Live: 126</p> <p>Protocol: UDP (17)</p> <p>Header Checksum: 0x216a [validation disabled]</p> <p>[Header checksum status: Unverified]</p> <p>Source Address: 128.238.29.23</p> <p>Destination Address: 128.238.38.160</p> <p>[Stream index: 1]</p> <p>User Datagram Protocol, Src Port: 53, Dst Port: 3163</p> <p>Domain Name System (response)</p> | <pre> 0000 00 09 6b 10 60 99 00 b0 8e 83 e4 54 08 00 45 00  ..k....T..E 0010 00 5a 45 95 00 00 7e 11 21 6a 80 ee 1d 17 80 ee  :?....&amp; 0020 26 a0 00 35 0c 5b 00 46 b0 ba 00 6e 81 00 00 01  &amp;5[F..n... 0030 00 02 00 00 00 00 03 77 77 77 04 69 65 74 66 03  ....w ww.ietf 0040 6f 72 67 00 00 01 00 01 c0 0c 00 01 00 01 00 00  org..... 0050 06 8e 00 04 84 97 06 4b c0 0c 00 01 00 01 00 00  ....K..... 0060 06 8e 00 04 41 f6 ff 33                          ....A..3 </pre> |
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Specifies if this is an individual (unicast) or group (broadcast/multicast) address (eth.dst.ig). 1 bit

Packets: 92 · Displayed: 2 (2.2%)

Profile: Default

## • Detail Type:

| No. | Time     | Source         | Destination    | Protocol | Length | Info                                                                         |
|-----|----------|----------------|----------------|----------|--------|------------------------------------------------------------------------------|
| 8   | 3.075845 | 128.238.38.160 | 128.238.29.23  | DNS      | 72     | Standard query 0x006e A www.ietf.org                                         |
| 9   | 3.076689 | 128.238.29.23  | 128.238.38.160 | DNS      | 104    | Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.255.51 |

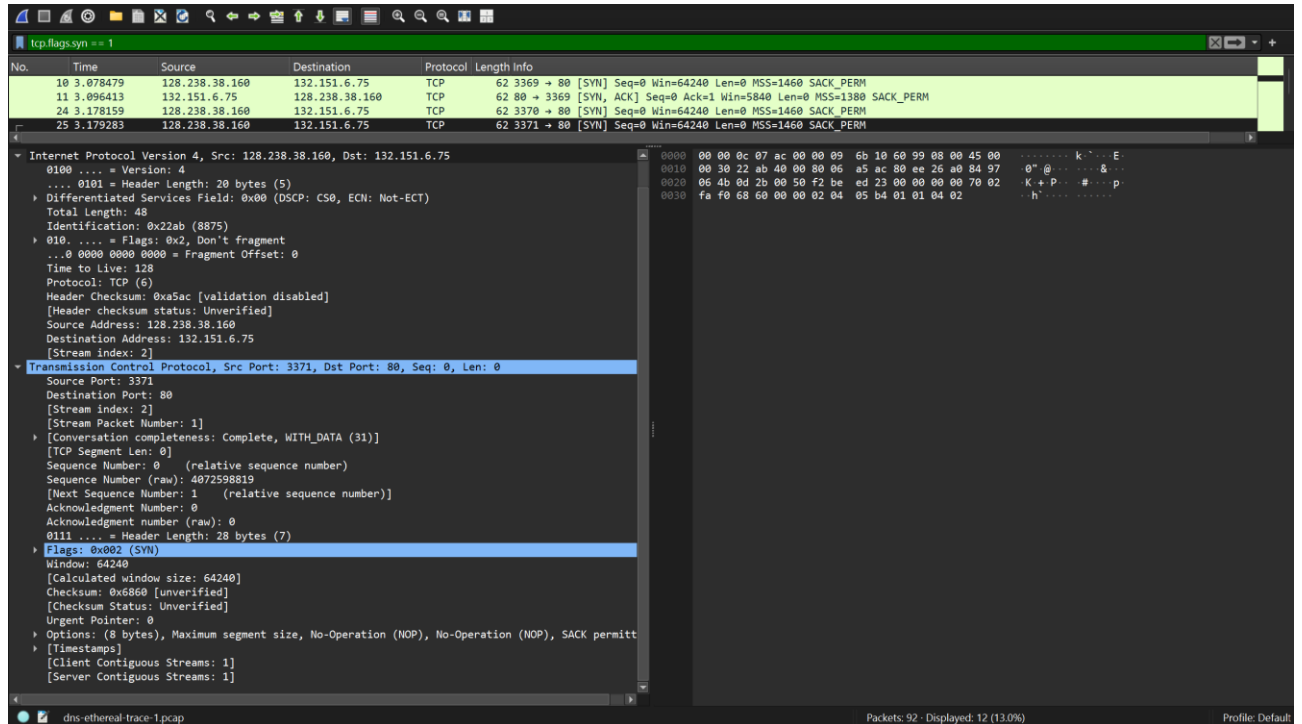
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Frame 8: Packet, 72 bytes on wire (576 bits), 72 bytes captured (576 bits)</p> <p>Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)</p> <p>Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23</p> <p>User Datagram Protocol, Src Port: 3163, Dst Port: 53</p> <p>Domain Name System (query)</p> <p>Transaction ID: 0x006e</p> <p>Flags: 0x0100 Standard query</p> <p>0... .. = Response: Message is a query</p> <p>000 0... .. = Opcode: Standard query (0)</p> <p>.... 0... .. = Truncated: Message is not truncated</p> <p>.... 1... .. = Recursion desired: Do query recursively</p> <p>.... 0... .. = Z: reserved (0)</p> <p>.... 0... .. = Non-authenticated data: Unacceptable</p> <p>Questions: 1</p> <p>Answer RRs: 0</p> <p>Authority RRs: 0</p> <p>Additional RRs: 0</p> <p>Queries</p> <p>www.ietf.org: type A, class IN</p> <p>[Response In: 9]</p> | <pre> 0000 00 00 0c 07 ac 00 00 09 6b 10 60 99 08 00 45 00  ....k....E 0010 00 3a 22 9e 00 00 00 11 d2 51 80 ee 25 a0 80 ee  :?....&amp; 0020 1d 17 8c 5b 00 35 00 26 8a cb 00 6e 01 00 00 01  [5&amp;...n... 0030 00 00 00 00 00 00 03 77 77 77 04 69 65 74 66 03  ....ww.ietf 0040 6f 72 67 00 00 01 00 01                          org..... </pre> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Identification of transaction (dns.id). 2 bytes

Packets: 92 · Displayed: 2 (2.2%)

Profile: Default

- **TCP SYN:**



## Bagian C – Soal 1–5

### Soal

1. Apa port tujuan pada pesan permintaan DNS? Apa port sumber pada pesan balasan DNS?
2. Ke alamat IP manakah pesan permintaan DNS dikirimkan? Apakah alamat IP tersebut merupakan default alamat IP server DNS lokal Anda?
3. Periksa pesan permintaan DNS. Apa "jenis" atau "type" dari pesan tersebut? Apakah pesan tersebut mengandung "jawaban" atau "answers"?
4. Periksa pesan balasan DNS. Berapa banyak "jawaban" atau "answers" yang terdapat di dalamnya. Apa saja isi yang terkandung dalam setiap jawaban tersebut?
5. Sertakan hasil tangkapan layar.

### Jawaban

#### 1. Port:

- Tujuan (request): 53
- Sumber (response): 53

#### 2. Alamat IP:

- Tujuan: 128.238.29.22
- Sumber: 128.238.38.160
- Keterangan: IP tujuan merupakan default server DNS lokal

#### 3. Type: A

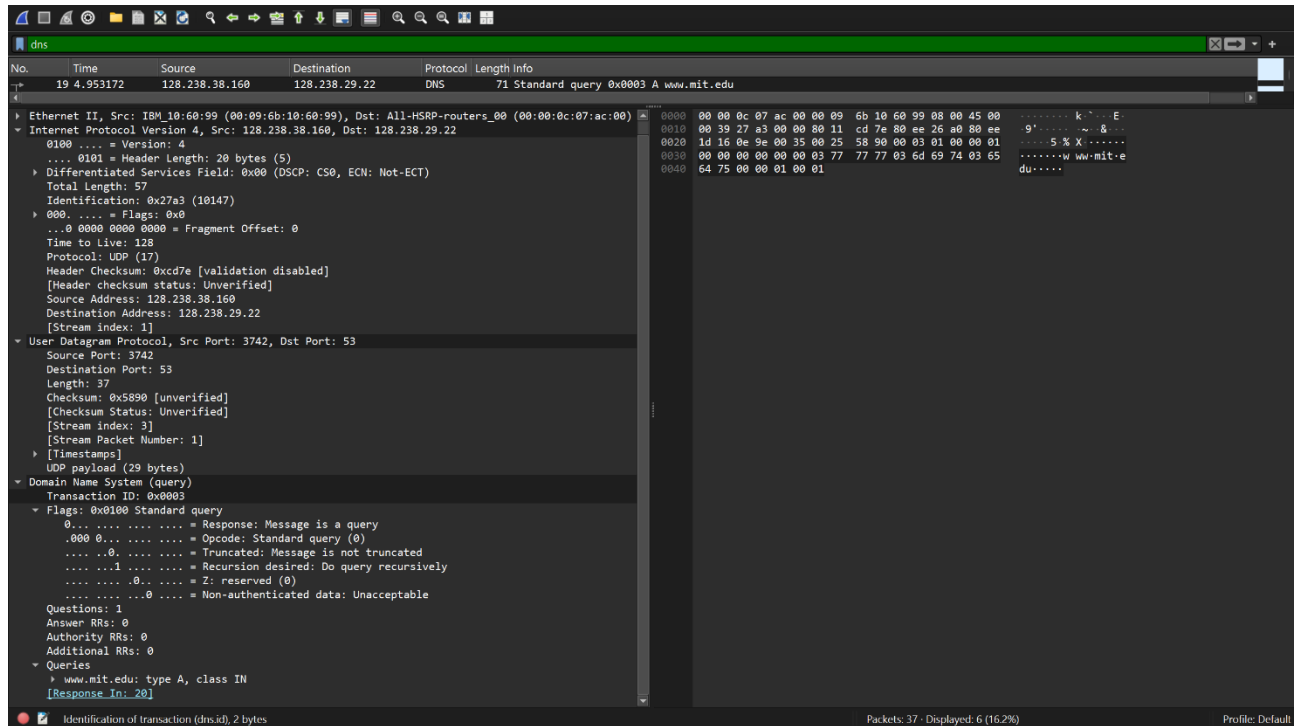
- Tidak ada *answers* pada request

#### 4. Balasan:

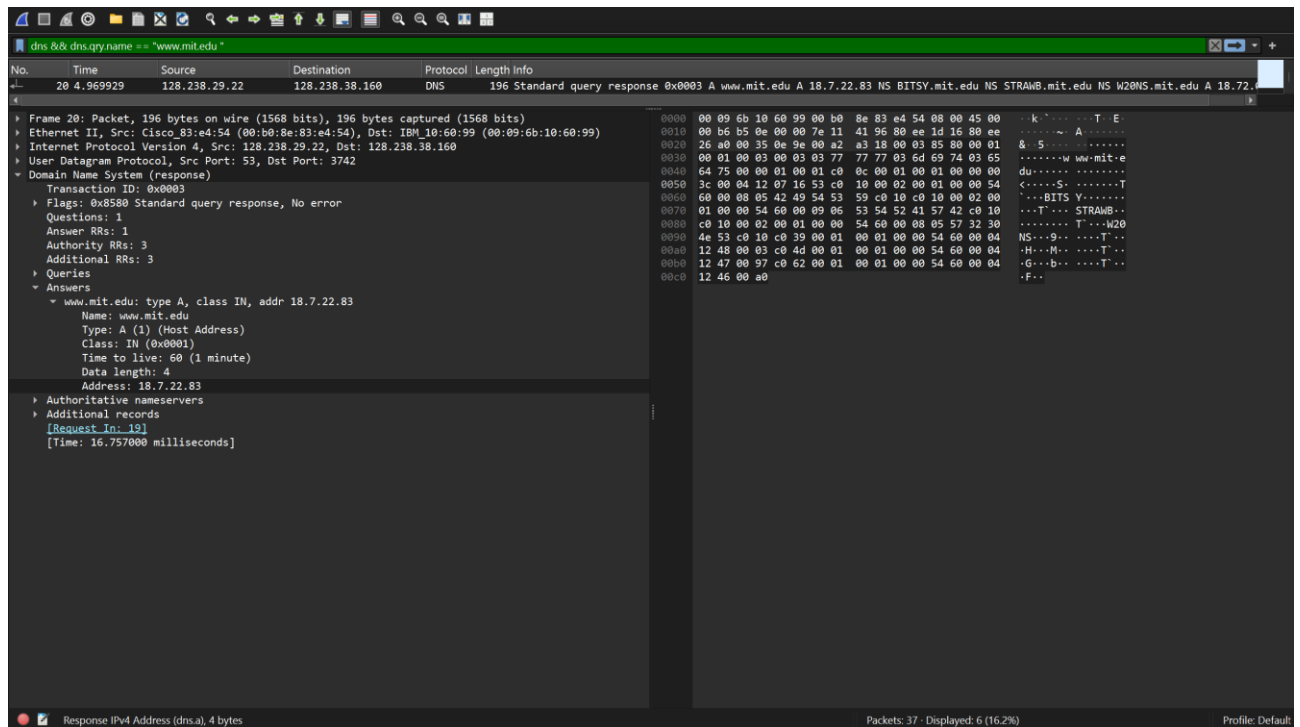
- Jumlah *answers*: 1
- Isi jawaban: alamat IP hasil resolusi untuk domain yang diminta

#### Bukti

- Request:



- Response:



## Bagian D – Soal 1–4

### 4.4 Analisis DNS – Bagian C (Soal 1–3)

#### Soal

1. Ke alamat IP manakah pesan permintaan DNS dikirimkan? Apakah alamat IP tersebut merupakan default alamat IP server DNS lokal Anda?
2. Periksa pesan permintaan DNS. Apa ”jenis” atau ”type” dari pesan tersebut? Apakah pesan tersebut mengandung ”jawaban” atau ”answers”?
3. Periksa pesan balasan DNS. Apa nama server MIT yang diberikan oleh pesan balasan? Apakah pesan balasan ini juga memberikan alamat IP untuk server MIT tersebut?
4. Sertakan hasil tangkapan layar.

#### Jawaban

##### 1. Alamat IP tujuan request:

- Request dikirim ke default DNS server: gpon.net
- IP server DNS lokal: fe80::1
- **Kesimpulan:** Ya, request dikirim ke default DNS server lokal

##### 2. Jenis query:

- Type: NS
- Request tidak mengandung *answers*

##### 3. Isi balasan DNS:

- Nama server MIT (Authority / nameservers):



- asia1.akam.net
- use2.akam.net
- ns1-37.akam.net
- ns1-173.akam.net
- eur5.akam.net
- use5.akam.net
- asia2.akam.net
- usw2.akam.net
- Balasan juga memberikan alamat IP beberapa server MIT:
  - eur5.akam.net → 23.74.25.64
  - use5.akam.net → 2.16.40.64
  - asia1.akam.net → 95.100.175.64
  - ns1-37.akam.net → 193.108.91.37
- Beberapa juga ada IPv6:
  - use5.akam.net → 2600:1403:a::40
  - ns1-37.akam.net → 2600:1401:2::25

---

## Bukti

### • Response:

```
Server: gpon.net
Address: fe80::1

Non-authoritative answer:
mit.edu nameserver = asia1.akam.net
mit.edu nameserver = use2.akam.net
mit.edu nameserver = ns1-37.akam.net
mit.edu nameserver = ns1-173.akam.net
mit.edu nameserver = eur5.akam.net
mit.edu nameserver = use5.akam.net
mit.edu nameserver = asia2.akam.net
mit.edu nameserver = usw2.akam.net

eur5.akam.net    internet address = 23.74.25.64
use5.akam.net    internet address = 2.16.40.64
asia1.akam.net   internet address = 95.100.175.64
ns1-37.akam.net  internet address = 193.108.91.37
use5.akam.net    AAAA IPv6 address = 2600:1403:a::40
ns1-37.akam.net  AAAA IPv6 address = 2600:1401:2::25
```

## Bagian E – Soal 1–4

### Soal

1. Ke alamat IP manakah pesan permintaan DNS dikirimkan? Apakah alamat IP tersebut merupakan default alamat IP server DNS lokal Anda?

2. Periksa pesan permintaan DNS. Apa "jenis" atau "type" dari pesan tersebut? Apakah pesan tersebut mengandung "jawaban" atau "answers"?
  3. Periksa pesan balasan DNS. Berapa banyak "jawaban" atau "answers" yang terdapat di dalamnya. Apa saja isi yang terkandung dalam setiap jawaban tersebut?
  4. Sertakan hasil tangkapan layar.
- 

**1. Alamat IP tujuan request:**

- Tujuan: bitsy.mit.edu → 18.0.72.3
- Bukan default server DNS lokal

**2. Jenis query:**

- Type: A
- Request **mengandung jawaban (answers)**

**3. Isi balasan DNS:**

- Alamat IP untuk www.aiit.or.kr:
    - 2606:4700:3031::ac43:9878
    - 2606:4700:3036::6815:4a08
    - 172.67.152.120
    - 104.21.74.8
- 

Bukti

- **Response:**

```
C:\Users\fathan>nslookup www.aiit.or.kr bitsy.mit.edu
Server:    bitsy.mit.edu
Address:   18.0.72.3

Non-authoritative answer:
Name:      www.aiit.or.kr
Addresses: 2606:4700:3031::ac43:9878
           2606:4700:3036::6815:4a08
           172.67.152.120
           104.21.74.8
```